Cell membrane

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Dr Iman

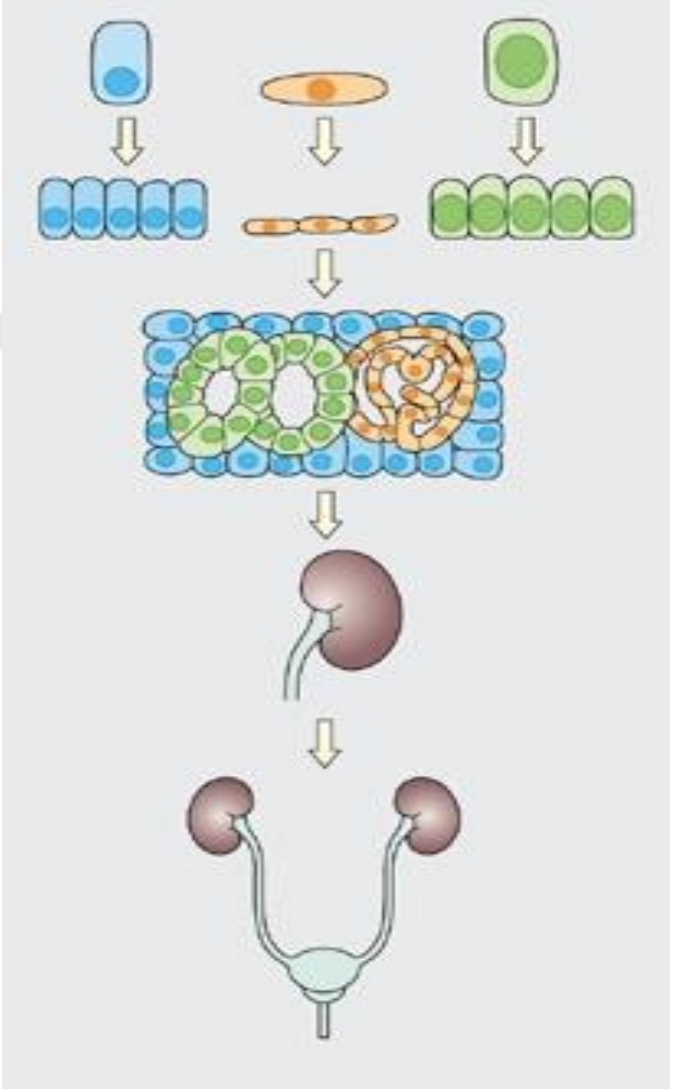

Why?
tell me
why!

- The cell is the functional and structural unit of the body.

- A group of cells with similar structure and function will form tissues.

-These tissues are grouped to form organs.

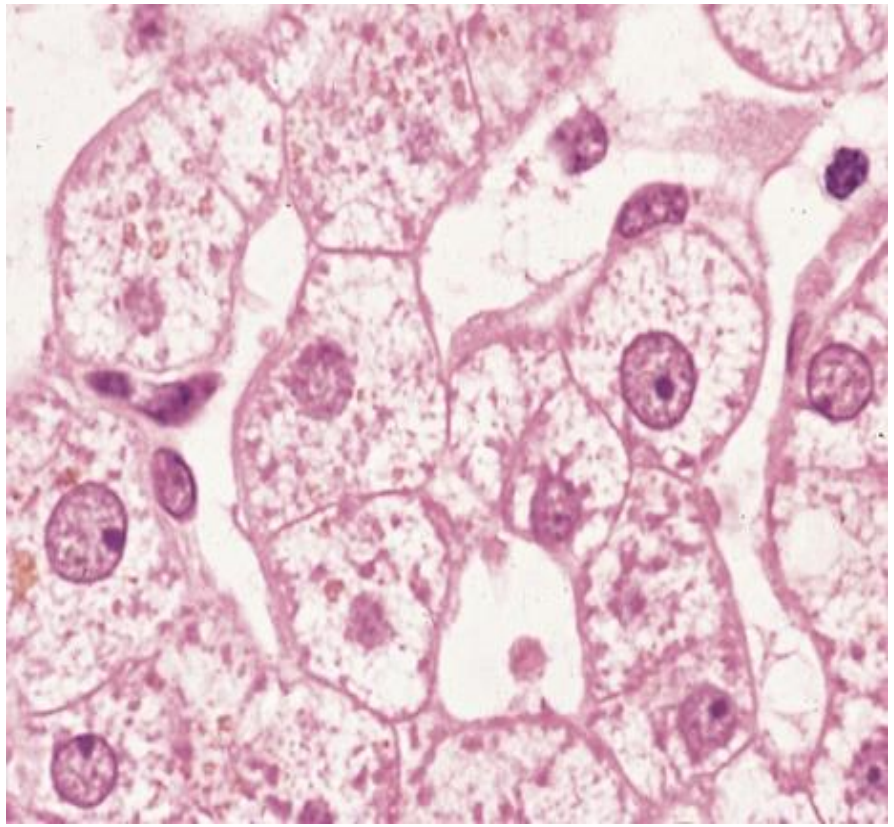
- A group of organs collect to form the body systems.



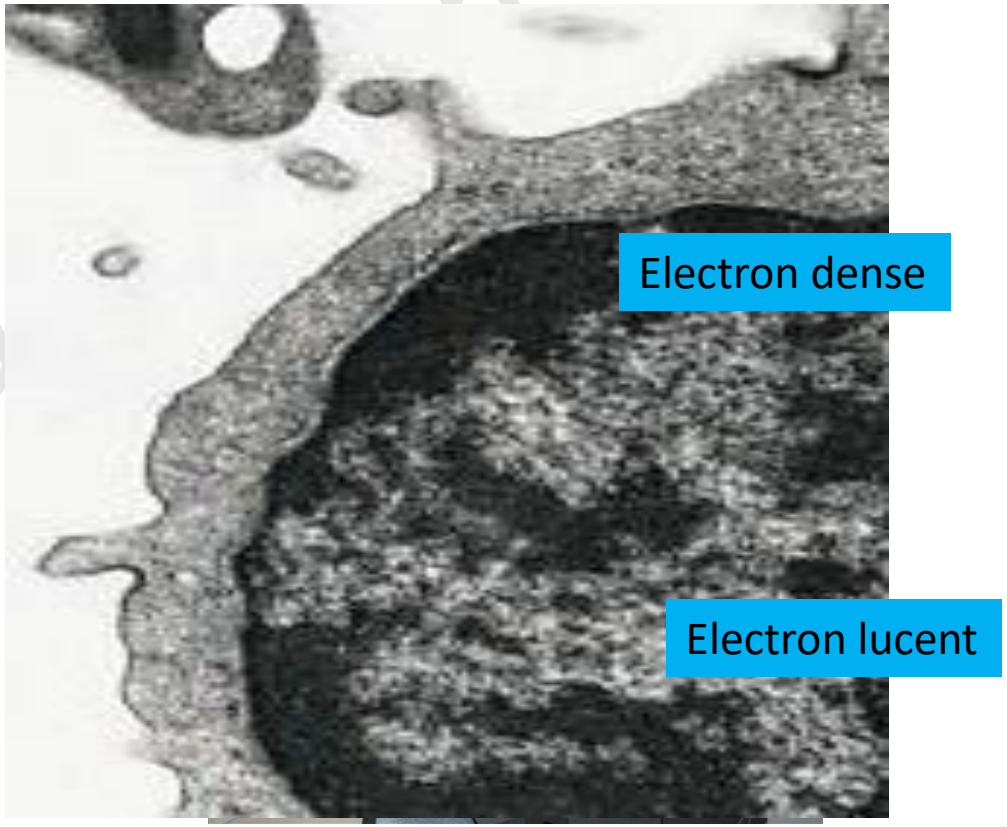
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Tools of studying Histology

LM



EM



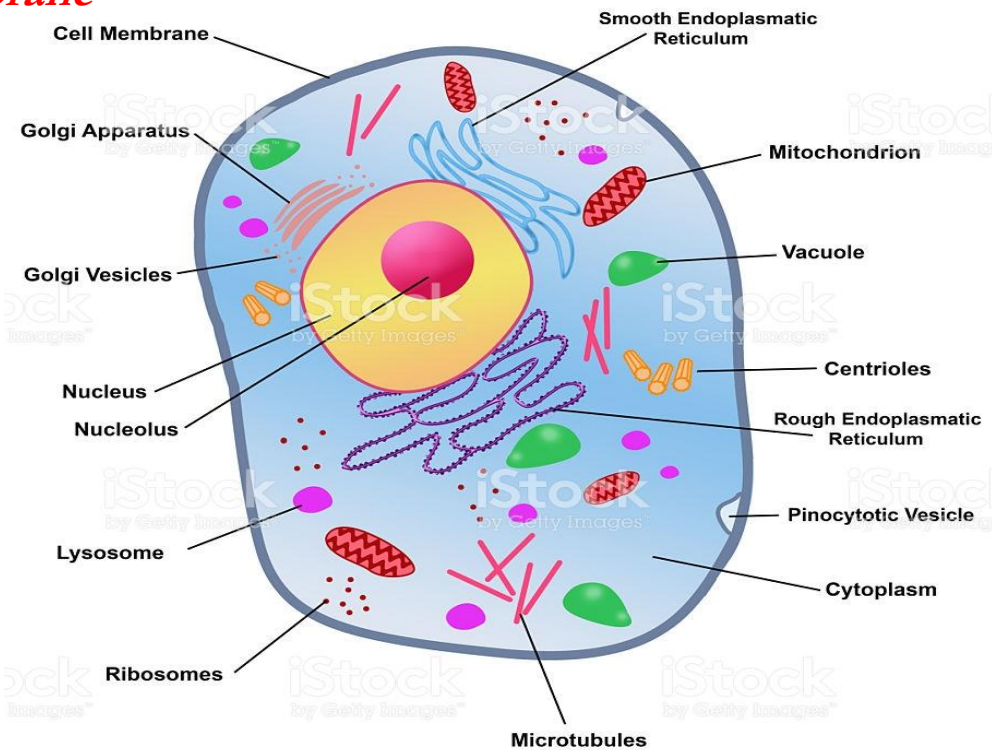
Light microscope

Electron microscope

ANIMAL CELL STRUCTURE

Cell membrane

Inclusions



Organelles

Cytoskeleton

Cell membrane

Plasma membrane = Plasmalemma

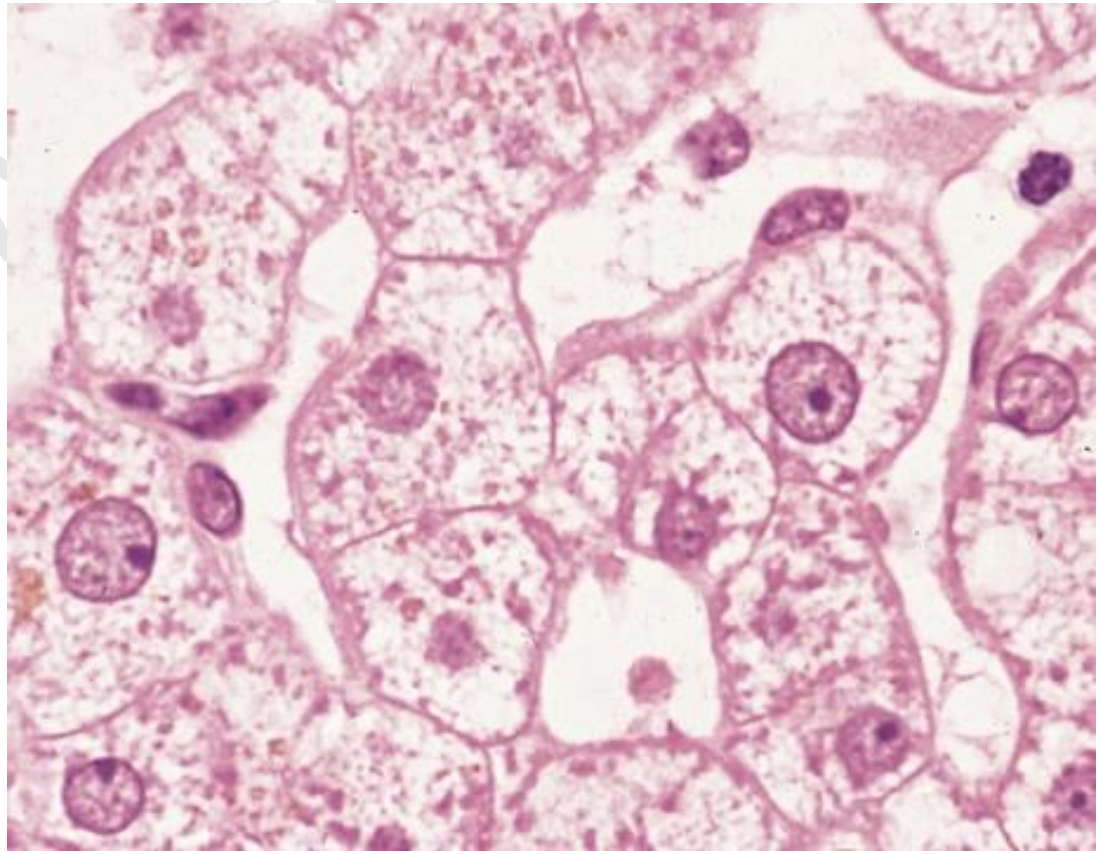
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Cell membrane

- LM
- EM
- Functions

LM

It can not be seen by light microscope *because* it is too thin to be seen *but* the condensation of the stain on the outer surface of the cell membrane marks its.



Dr Iman Al

EM

1- *At low magnification:* thin dense line 8 - l0 nm in thickness.

2- *With higher magnification:* a trilaminar structure, with an outer (= extracellular leaflet) and an inner (= cytoplasmic leaflet) electron dense lines and a middle electron lucent zone in between.

The entire structure is known as the unit membrane.



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Molecular structure of the Cell membrane

3 components:

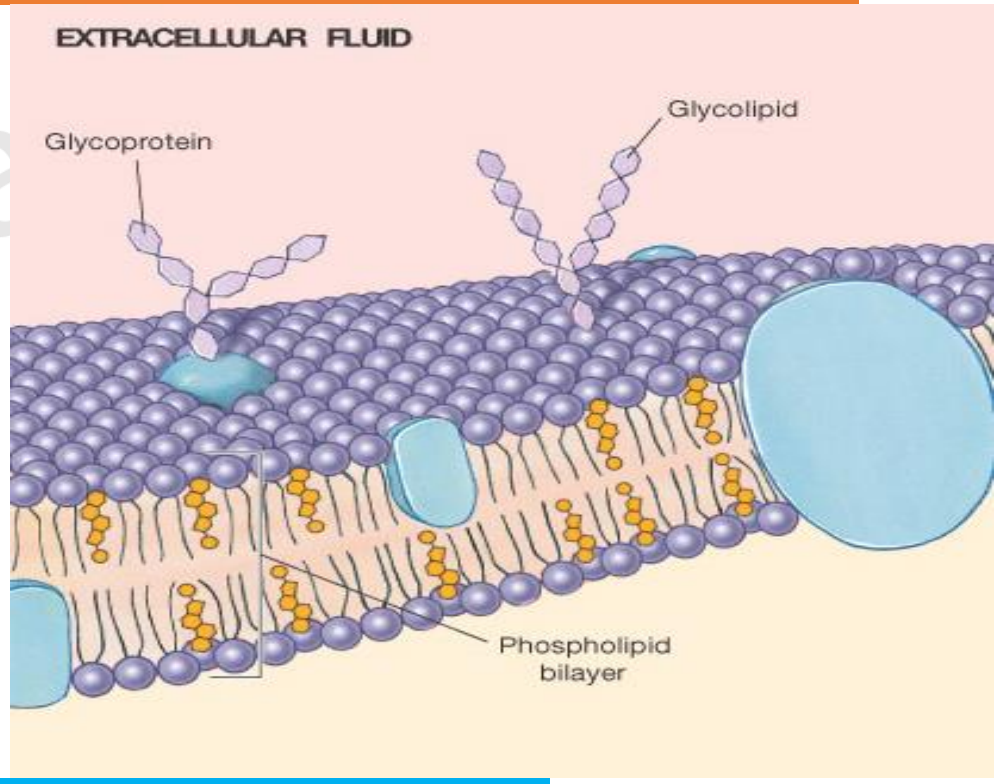
1-Lipid molecules:

a- phospholipids

b- cholesterol

2- Protein molecules: 50% of the membrane mass.

3- Carbohydrate molecules



The cell membrane and almost all the membranes surrounding the membranous organelles have the same structure except for minor differences.

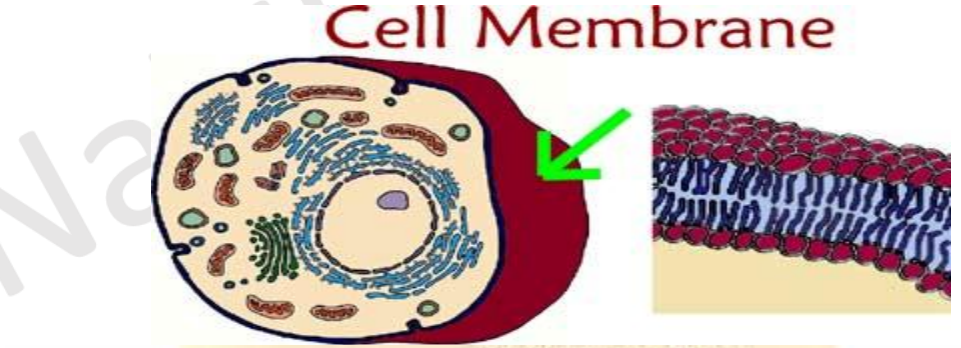
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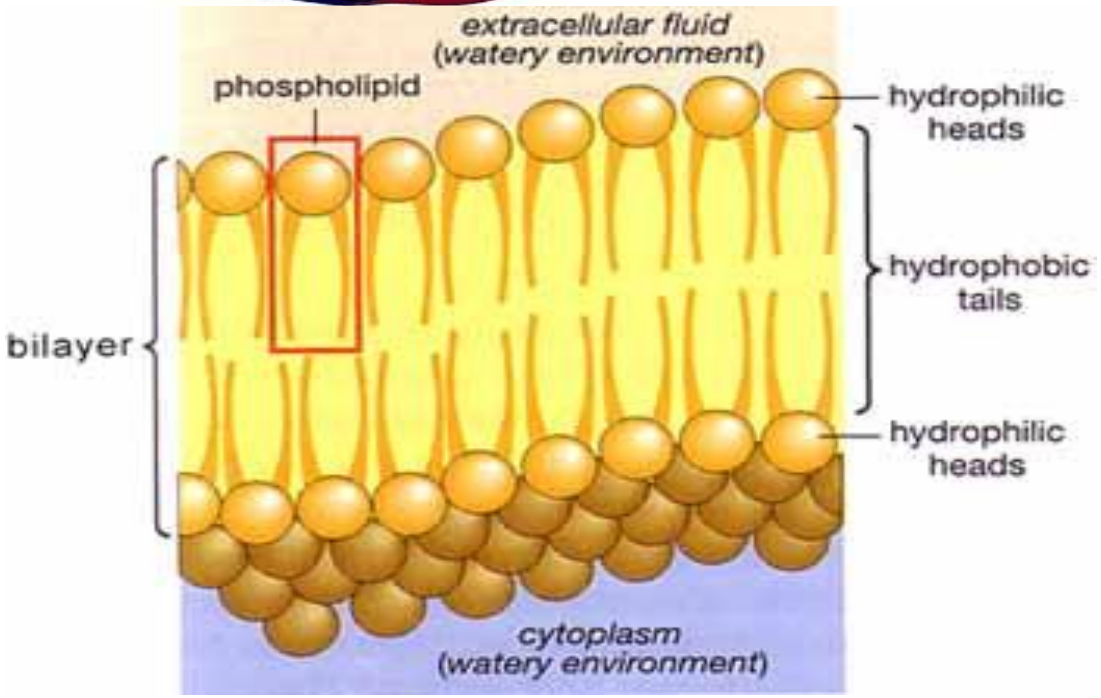
1- Lipid molecules
A-Phospholipids

- Each phospholipid molecule consists of:

1- One polar hydrophilic head : faces the aqueous media on either side of the membrane.

2- Two long non-polar hydrophobic tails (fatty acids): project towards the center of the membrane facing each other. They form weak non-covalent bonds with each other, holding the bilayer together.

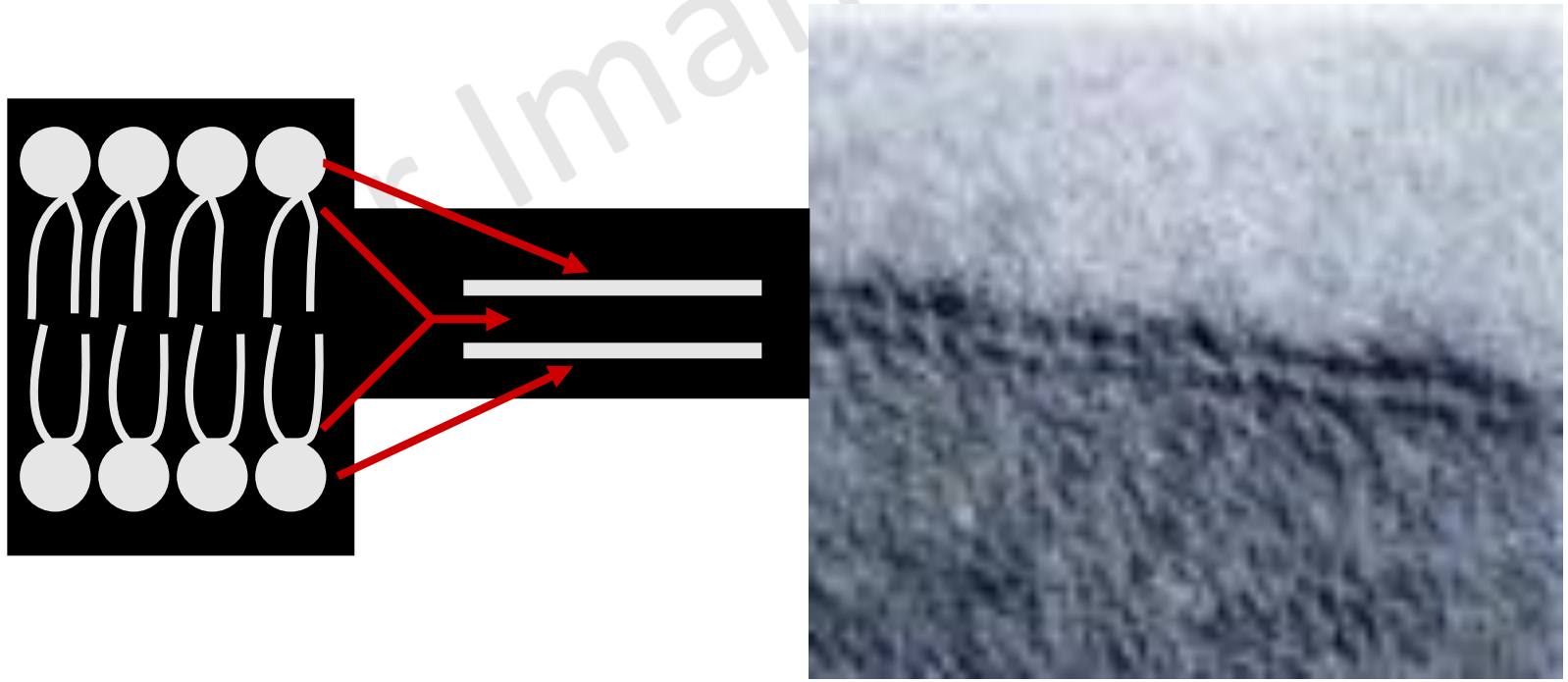




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Why does the cell membrane appear as a trilaminar structure?

Deposition of osmium in the hydrophilic heads



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Functions of phospholipid molecules

- Prevent passage of:

1-Water soluble substances.

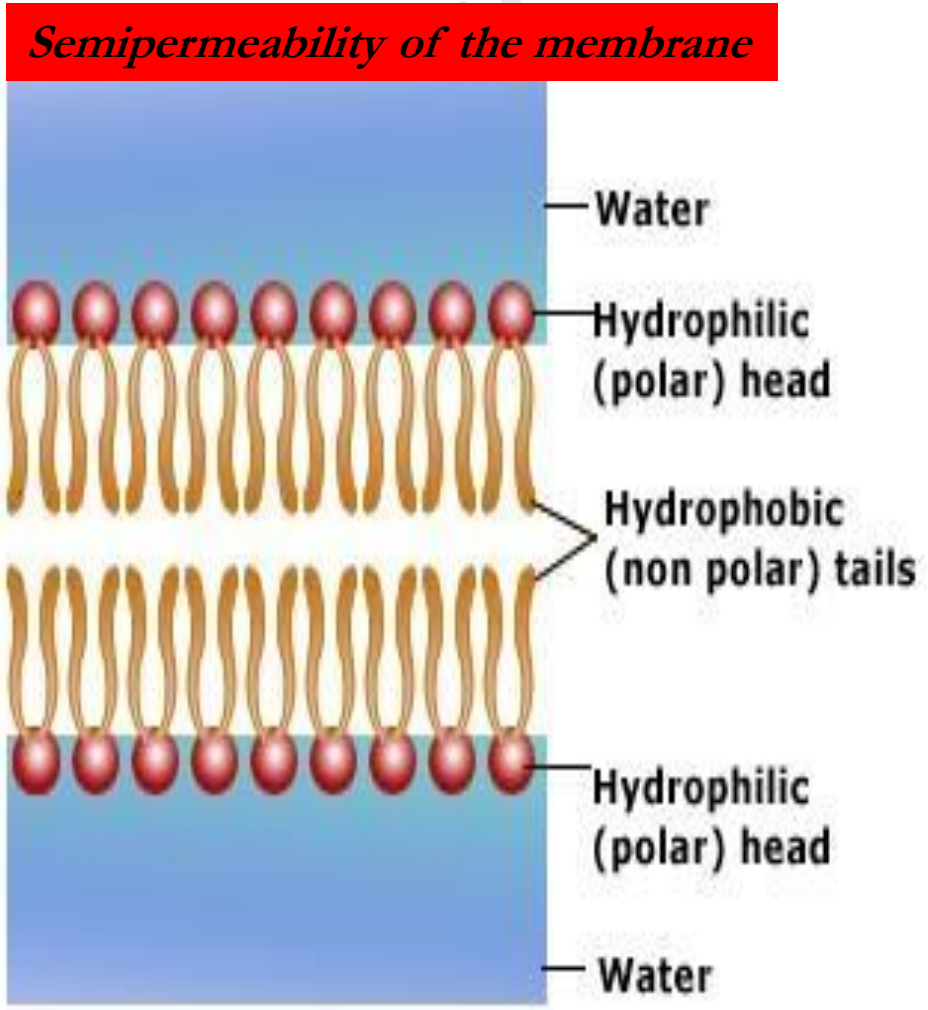
2- Polar ions.

- Allow passages of:

1- Fat soluble substances.

2- Non polar substances.



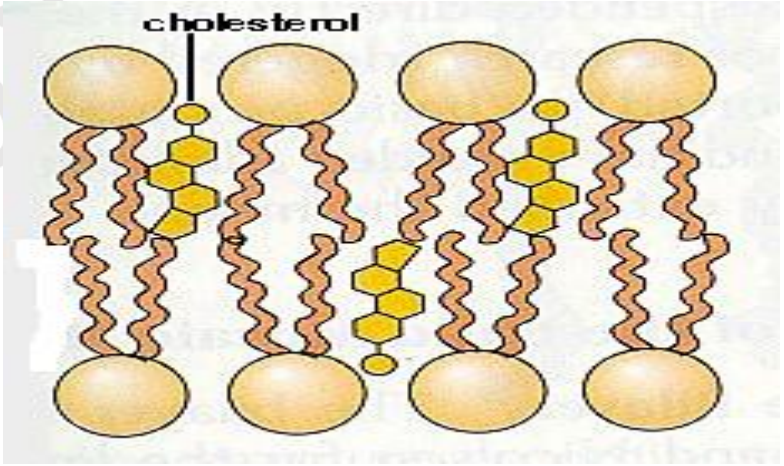


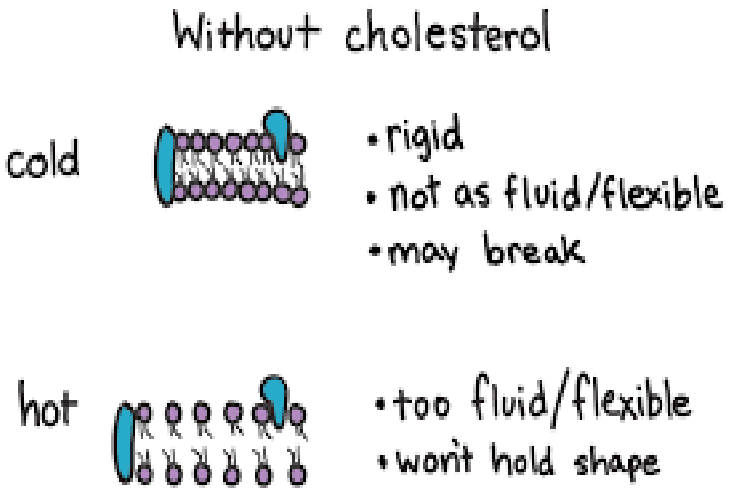
B-Cholesterol

- They are incorporated within the lipid bilayer.
- Functions:

1- Stability of the membrane.

2-Regulation of membrane fluidity in body temperature.





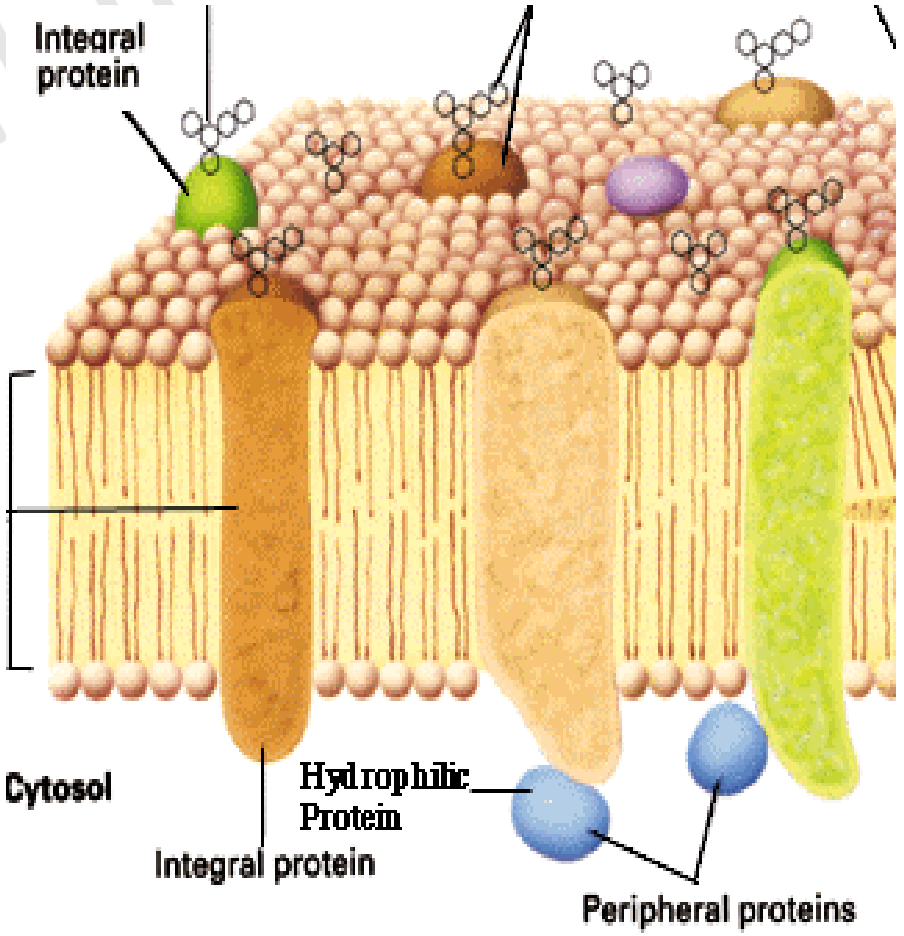
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2- Protein molecules

- Two types:

A-Integral membrane proteins:

- They are embedded within the lipid bilayer.
- Most of these proteins traverse the whole thickness of the membrane and are called *transmembrane proteins*.



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There are six functional forms of integral membrane proteins :

1- *Pumps* : transport ions (Na+, K+) actively across the membrane.

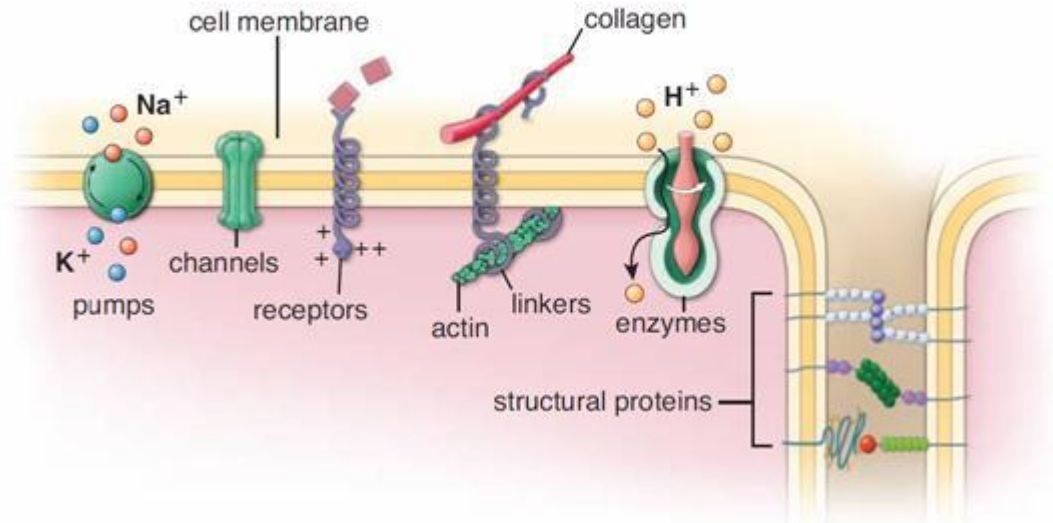
2- *Channels*: transport substances passively.

3- *Receptors*: allow binding of specific molecules e.g. hormone.

4- *Enzymes*: ATP synthase of the inner mitochondrial membrane and some types of digestive enzymes in the small intestine.

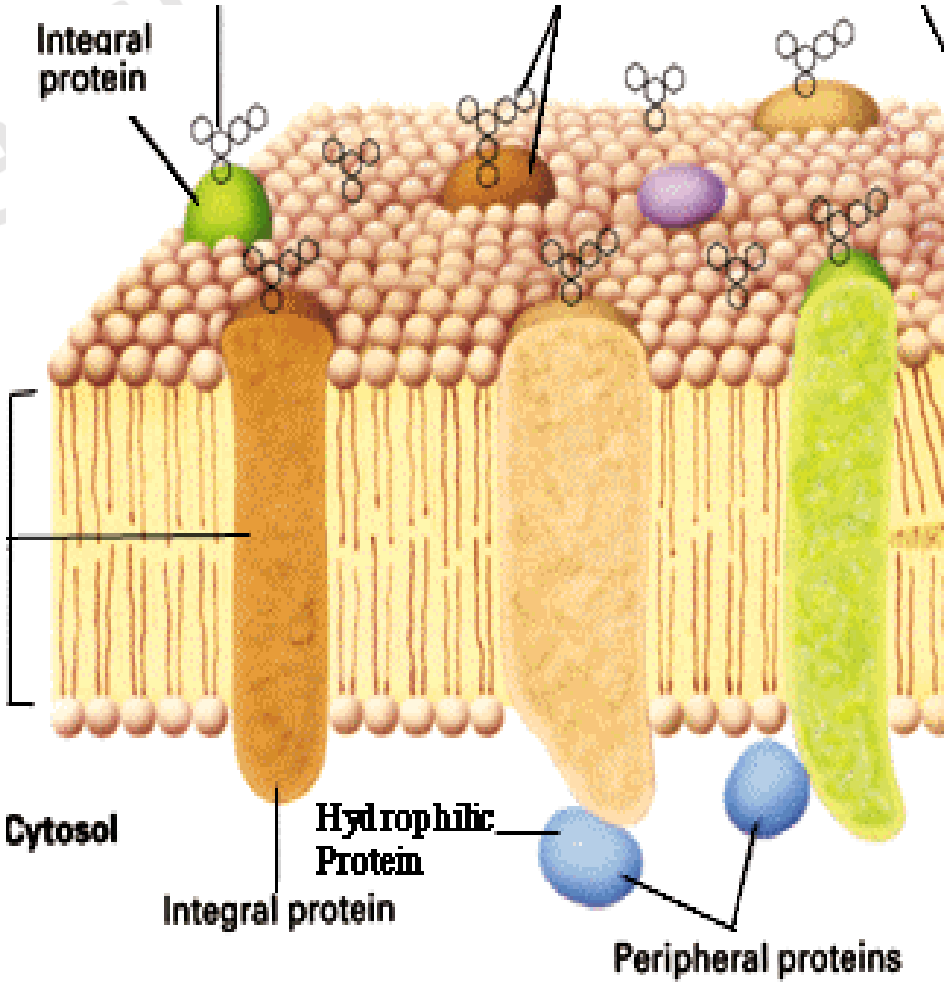
5- *Linkers* : anchor the intracellular cytoskeleton to the extracellular matrix.

6- *Structural proteins:* form junctions between neighboring cells.



B-Peripheral membrane proteins

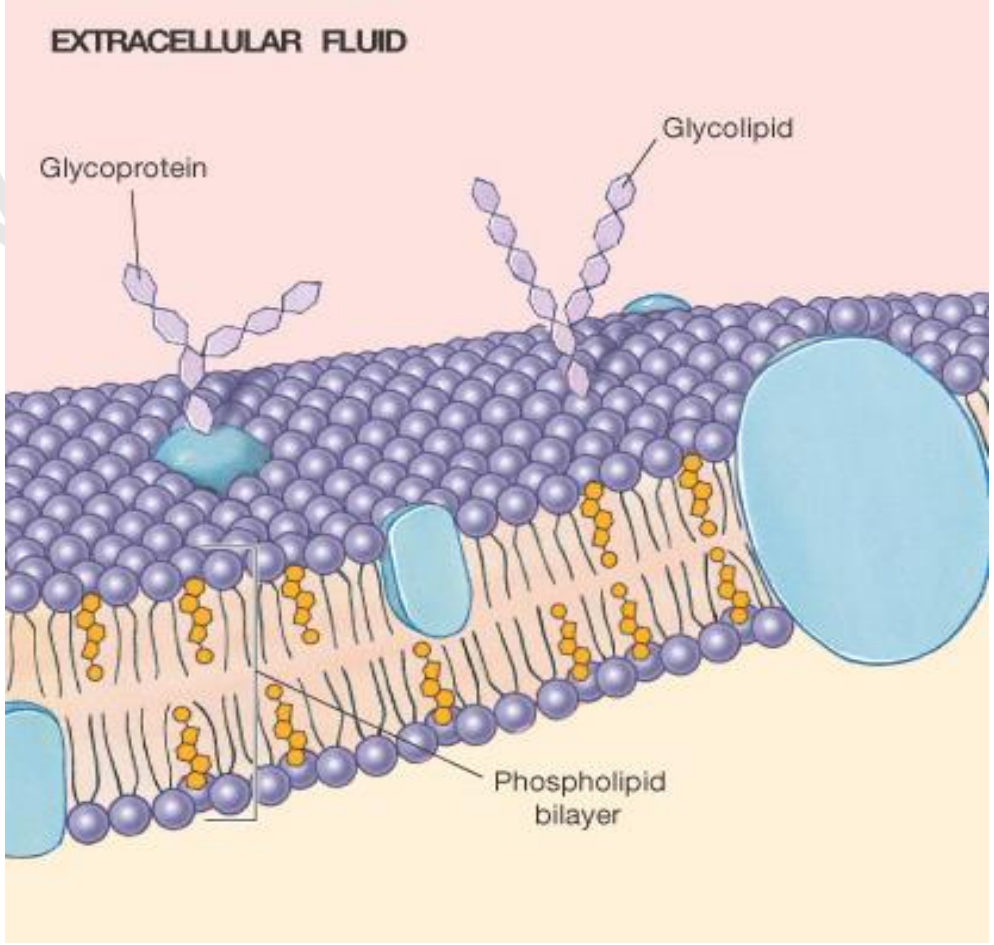
- They are not embedded into lipid bilayer but they are *loosely-associated* with membrane.
- They are usually located on the cytoplasmic surface of the membrane.
- Function: form a link between the cell membrane and the cytoplasmic components.



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3- Carbohydrate molecules

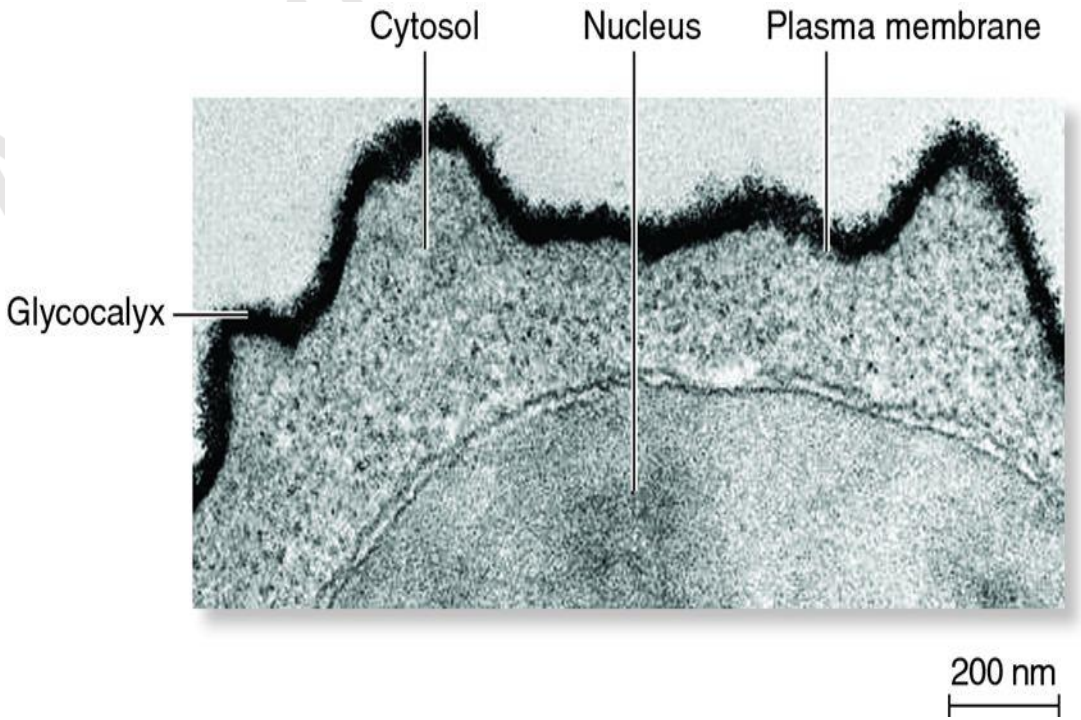
- They are present as glycoproteins and glycolipids of the cell membrane.
- They are oriented towards the outside of the membrane forming the ***cell coat*** or ***glycocalyx***.



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3- Carbohydrate molecules

- The cell coat is represented by the "*fuzzy*" material on the outer surface of the membrane.



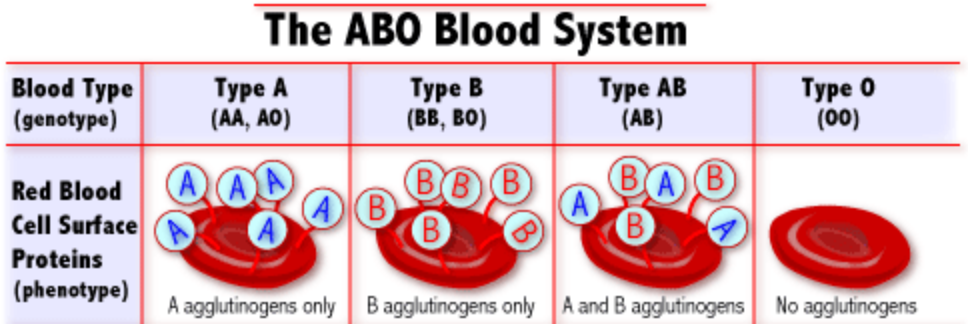
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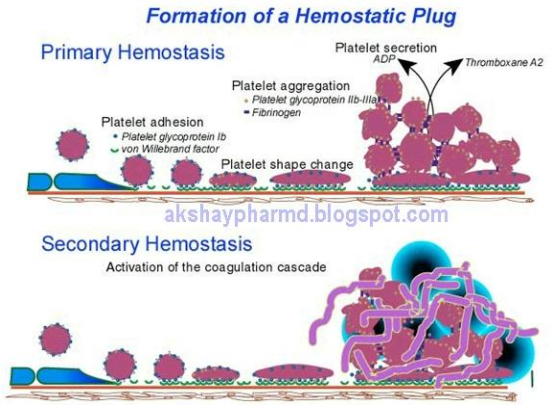
Functions of cell coat

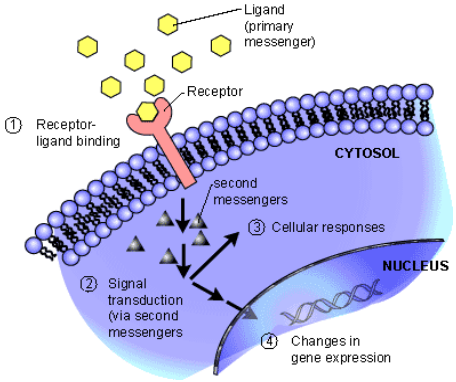
1- *Cellular recognition* e.g. the glycocalyx on the surface of red blood cells determines the four blood groups.

2- *Cell-cell adhesion*.

3- *Receptor:* for ligands by the glycoproteins of the cell membrane.



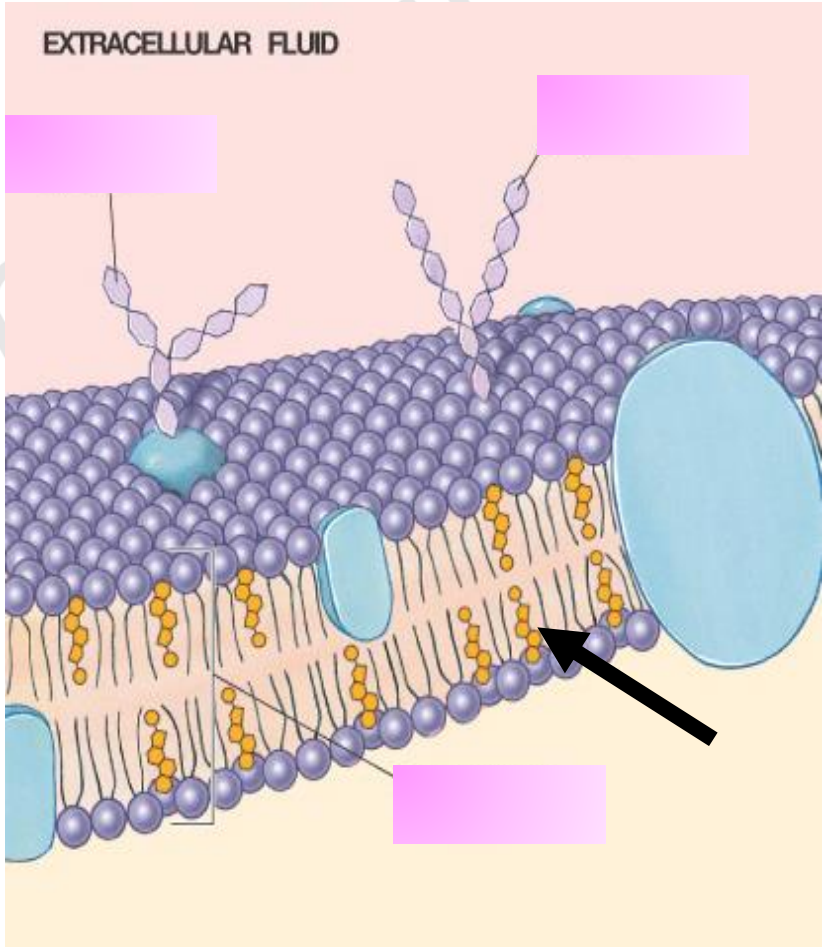




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What is the function of the pointed structure?

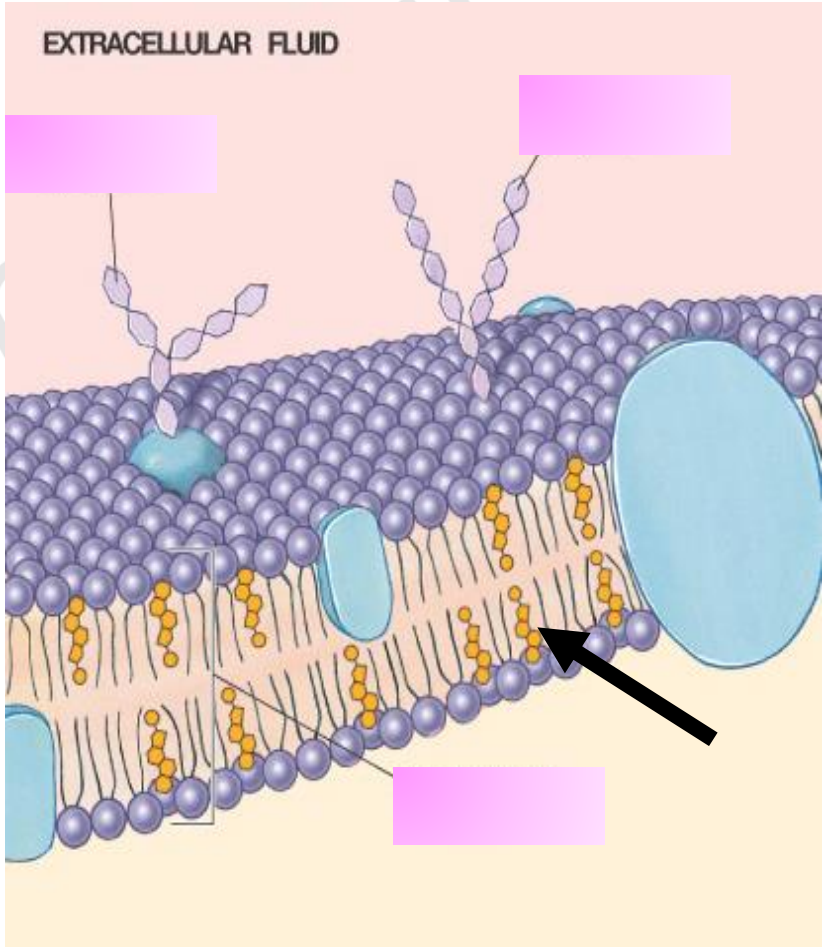
a. Acts as a channel.
b. Connects the cytoskeleton of the cell with its matrix.
c. Controls membrane fluidity.
d. Cell to cell adhesion.



Dr Imar

What is the function of the pointed structure?

a. Acts as a channel.
b. Connects the cytoskeleton of the cell with its matrix.
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Thank You